# Maximilian L. Schulten

AI/ML Engineer & Researcher

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## EDUCATION

**Case Western Reserve University** — **Cleveland, OH**

*M.S./B.S. Computer Science, B.S. Mathematics* — *08/2022 – Present (Exp. Summer '27)*

- Integrated B.S./M.S. Candidate
- Cumulative GPA: 3.91, Dean's High Honor List (all semesters)
- Teaching assistant for CSDS 302 (Discrete Math) & CSDS 101 (Comp. Sci. and Data Sci. for all)
- Member of Dr. Sumon Biswas' reSAID Lab – LLM-generated code security evaluation; knowledge distillation for SWE tasks

## EXPERIENCE

**Medpace, Inc** — **Cincinnati, OH**

*Clinical Informatics Intern* — *05/2026 – 08/2026*

- Built a print order platform in R Shiny on top of the Microsoft Graph REST API, using optimistic UI updates and async programming to keep the interface responsive under concurrent load
  - Serves $\approx 50$ orders per week across 200+ daily sessions; available company-wide
- Built internal auditing tool used to parse the codebases of published internal applications for vulnerabilities through the GitHub and Posit Connect APIs
  - Used to supervise the transition of $50+$ applications to CI via Git-backed deployment
  - Leveraged the R abstract syntax tree to extract hardcoded DSNs and calls to unverified data sources
  - Implemented an asynchronous and concurrent FIFO audit queue using R Shiny module servers and extended tasks to prevent blocking behavior

**Circana, Inc** — **Chicago, IL**

*LiquidAI Strategy Intern* — *06/2025 – 08/2025*

- Structured AI-backed data visualizations and actionable insights from Circana proprietary POS and market data, built with JavaScript, AmCharts4, and native JSON parsing
- Engineered automated visualization pipelines that moved reporting to a 24-hour refresh cadence, reducing manual effort and keeping deliverables current

**Medpace, Inc** — **Cincinnati, OH**

*Informatics Co-Op Intern* — *01/2025 – 05/2025*

- Fitted Implemented Isolation Forest model to identify outlier clinical data; developed interactive web application integrated with a Snowflake database to perform real-time anomaly predictions
- Built non-linear regression models for site startup timelines, improving forecast accuracy by $\approx 18\%$ over legacy methods
- Revamped internal feasibility tool; replaced local spreadsheet system with a database-backed, full-stack application using MS SQL Server and R Shiny in cooperation with analyst
- Scraped and visualized public EudraCT trial data using Python to analyze competing studies

## PROJECTS

*LeftForTheReader (L4TR)* — *2025*

- LLM chat interface for math using `llama.cpp` and Streamlit; performs RAG using more than 30k proofs/definitions scraped from ProofWiki using BS4 and stored in ChromaDB; deployed on hardened VPS via Docker

*SkillSync* — *2026*

- Senior project matching resumes to jobs; wrote NLP and ML pipeline; utilizes SVM with embedding model features to classify resumes into industries; leverages transformer models for zero-shot NER via GLiNER; CI/CD tests using GitHub Actions

*Image Compression via Learned Basis Representations* — *2026*

- Implemented DCT/JPEG, PCA, and ICA compression from scratch in Python/NumPy; evaluated RMSE, PSNR, and MSSIM across methods on BSD68 and Kodak datasets; improved PCA compression ratio 4× via integer quantization with component-wise whitening

*Causal Structure Preservation in Tabular Synthetic Data Gen.* — *2026*

- Evaluated causal fidelity of TabDDPM, CTGAN, and VAE across 7 datasets using PC, FCI, LiNGAM, and NOTEARS; found statistical quality and causal preservation to be unrelated

## SKILLS

**ML/AI** HuggingFace, vLLM, PyTorch, scikit-learn, SLURM, NumPy, Pandas, TRL

**Software** Python, TS, Java, Flask, SQL, Git, R, R Shiny, Docker, C